First-Order-Logic

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1. Basic Statements (Properties)

These use predicates to describe characteristics of an object.

- **Statement:** "Tweety is a bird."
  - FOL: $Bird(Tweety)$
- **Statement:** "Someone is happy."
  - FOL: $\exists x \, Happy(x)$
- **Statement:** "Not everyone is tall."
  - FOL: $\neg \forall x \, Tall(x)$

2. Universal Quantifier (∀) Examples (All/Every)

- **Statement:** "All cats are mammals."
  - FOL: $\forall x\ (Cat(x) \rightarrow Mammal(x))$
- **Statement:** "Every student loves AI."
  - FOL: $\forall x\ (Student(x) \rightarrow Loves(x, AI))$
- **Statement:** "All flowers are beautiful."
  - FOL: $\forall x\ (Flower(x) \rightarrow Beautiful(x))$

3. Existential Quantifier (∃) Examples (Some/There is)

- **Statement:** "Some people are honest."
  - FOL: $\exists x \, (Person(x) \land Honest(x))$
- **Statement:** "There is a king who is wise."
  - FOL: $\exists x \, (King(x) \land Wise(x))$
- **Statement:** "Some students do not like homework."
  - FOL: $\exists x \, (Student(x) \land \neg Likes(x, Homework))$

4. Combined Quantifiers (Complex Statements)

- **Statement:** "Every person has a mother." (Usually, this implies one specific mother per person, but in basic FOL, we often interpret it as "there exists at least one mother").
  - **FOL:** $\forall x\ (Person(x) \rightarrow \exists y\ (Person(y) \land Mother(y, x)))$
- **Statement:** "Everyone loves someone." (Different people might love different people).
  - **FOL:** $\forall x\ (Person(x) \rightarrow \exists y\ (Person(y) \land Loves(x, y)))$
- **Statement:** "There is someone whom everyone loves." (There is one specific person that everyone loves).
  - **FOL:** $\exists y\ (Person(y) \land \forall x\ (Person(x) \rightarrow Loves(x, y)))$

Thank you.